
Funktionentheorie

Autumn Semester 2026

Lecture Notes

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References

[1] Imamoglu, Özlem: Vorlesungsnotizen Funktionentheorie. ETH Zürich. 2026

Funktionentheorie

Fynn Krebsler—fkrebsler@student.ethz.ch

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1 Introdcution

1.1 Motivation

Consider the following example. On $\mathbb{R}$ it is not too difficult to construct a function f that is differentiable n times but not differentiable $n+1$ times. For example,

$$f(x) = \begin{cases} x^2 \sin(\frac{1}{x^2}) & x \neq 0, \\ 0 & x = 0 \end{cases}.$$

This function is differentiable but the derivative is not continuous. Integrating, we find a function that is differentiable twice but not three times. In fact, we can construct a function that is differentiable n times but not $n+1$ times for any $n \in \mathbb{N}$ using this method.

In contrast, we'll see that if $f : \mathbb{C} \rightarrow \mathbb{C}$ is differentiable once, then it is differentiable infinitely many times.

Another example is that for real numbers, there are functions $f : \mathbb{R} \rightarrow \mathbb{R}$ which are infinitely differentiable but not analytic. I.e. the Taylor series at some point x_0 exists, but does not converge to the function in any neighborhood of x_0 . For example,

$$x \mapsto (x) = \begin{cases} e^{-\frac{1}{x^2}} & x \neq 0, \\ 0 & x = 0 \end{cases},$$

where all derivatives are zero so the Taylor series is identically zero.

In contrast, we'll see that a differentiable function $f : \mathbb{C} \rightarrow \mathbb{C}$ has a power series representation. So differentiability is equivalent to analyticity.

A further comparison is that there are many examples of functions $f : \mathbb{R} \rightarrow \mathbb{R}$ which are C^∞ and bounded, for example

$$f(x) = \sin(x).$$

In contrast, we'll see that if $f : \mathbb{C} \rightarrow \mathbb{C}$ is bounded and differentiable, then it is constant.

The robustness of this theory can be used for applications which seem to have nothing to do with complex analysis. Probably the most used examples are evaluating real integrals.

$$\int_0^\infty \cos(t^2) dt = \int_0^\infty \sin(t^2) dt = \frac{\sqrt{2\pi}}{4}.$$

Another famous example is the prime number theorem, which states that the number of primes less than x is asymptotic to $\frac{x}{\log(x)}$.

Another classic is to consider a non-zero polynomial $f(x) \in \mathbb{C}[x]$. Then the Fundamental Theorem of Algebra states that f has a zero in $\mathbb{C}$.

Lastly, consider

$$r_4(n) = \#\{(x, y, z, w) \in \mathbb{Z}^4 : x^2 + y^2 + z^2 + w^2 = n\}.$$

Lagrange proved that every positive integer has such a representation, namely $r_4(n) \geq 1$. Jacobi later proved that

$$r_4(n) = 8 \sum_{d|n; 4 \nmid d} d.$$

1.2 Review of Complex Numbers

Recall that the complex numbers are given as

$$\mathbb{C} = \{x + iy : x, y \in \mathbb{R}\},$$

where $i^2 = -1$. We can embed $\mathbb{R}$ into $\mathbb{C}$ via $x \mapsto x + i0$. We also defined

Definition 1.1: Real and Imaginary Parts

For $z = x + iy \in \mathbb{C}$, we define the **REAL PART** $\Re(z) = x$ and the **IMAGINARY PART** $\Im(z) = y$.

Furthermore, the **COMPLEX CONJUGATE** is defined as

$$\bar{z} = x - iy.$$

From this it follows that

Lemma 1.2:

For $z = x + iy \in \mathbb{C}$, we have

$$z \in \mathbb{R} \Leftrightarrow z = \bar{z} \quad z \in i\mathbb{R} \Leftrightarrow z = -\bar{z}.$$

$\mathbb{C}$ can also be represented by pairs of real numbers in $\mathbb{R}^2$. For example $z = (x, y), w = (u, v) \in \mathbb{R}^2$ where

$$z = w \Leftrightarrow x = u, y = v.$$

Addition can be defined in both notations as

$$z + w = (x + u, y + v) \quad z + w = (x + u) + i(y + v).$$

Multiplication was a bit more cumbersome. Using the definition of i , we have

$$\begin{aligned} z \cdot w &= (x + iy)(u + iv) \\ &= xu + i^2 yv + i(xv + yu) \\ &= (xu - yv) + i(xv + yu). \end{aligned}$$

So in the pair representation, multiplication is given by

$$(x, y) \cdot (u, v) = (xu - yv, xv + yu).$$

Note that in the pair representation,

$$i \leftrightarrow (0, 1) \quad \text{and} \quad 1 \leftrightarrow (1, 0).$$

The good thing about the pair notation, $(\mathbb{R}^2, +, \cdot)$ satisfies

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1. $(\mathbb{R}^2, +)$ is an abelian group with identity $(0, 0)$
2. $(\mathbb{R}^2 \setminus \{(0, 0)\}, \cdot)$ is an abelian group with identity $(1, 0)$
3. The distributive law holds: $z_1(z_2 + z_3) = z_1z_2 + z_1z_3$

Together, these properties make $(\mathbb{R}^2, +, \cdot)$ a field. So the complex numbers with the above addition and multiplication form a field.

The additive inverse of $z = (x, y)$ is given by $-z = (-x, -y)$. The multiplicative inverse of $0 \neq z = (x, y)$ is given by

$$z^{-1} = \frac{\bar{z}}{|z|^2} = \frac{(x, -y)}{x^2 + y^2} = \left(\frac{x}{x^2 + y^2}, -\frac{y}{x^2 + y^2} \right).$$

Here, $|z| \in \mathbb{R}$ is the **MODULUS** or **NORM** of z , defined as

$$|z| = \sqrt{x^2 + y^2}.$$

We can also consider the **POLAR COORDINATES** of z . Any point on the plane $\mathbb{R}^2$ can be represented by a distance r from the origin and an angle θ from the positive x -axis. Here,

$$z = r(\cos(\theta) + i \sin(\theta)) = re^{i\theta}.$$

Beware that the representation is not unique, since θ can be replaced by $\theta + 2\pi k$ for any $k \in \mathbb{Z}$, unless we restrict θ to a certain interval. The standard is to take $\theta \in (-\pi, \pi]$. θ is called the **ARGUMENT** of z , denoted $\arg(z)$, and r is the modulus $|z|$. For the argument in $(-\pi, \pi]$, we write $\text{Arg}(z)$.

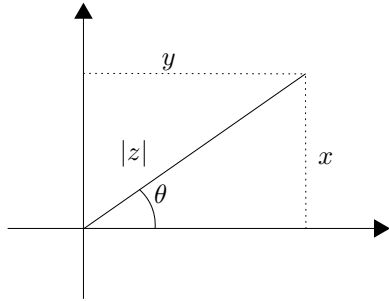


Figure 1: Polar coordinates of a complex number z

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